

The complete guide to open-source LLM evaluation in 2026

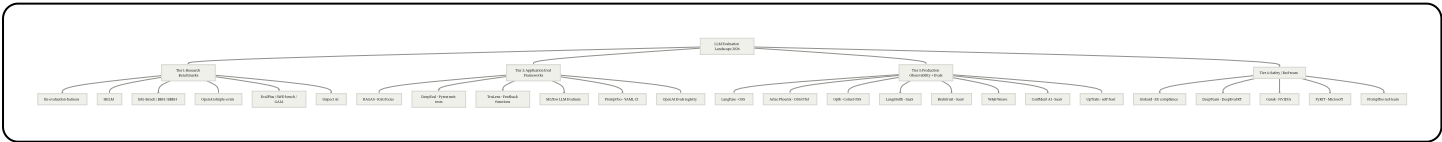
LLM evaluation has matured from academic leaderboards into a production discipline with **three distinct layers: reproducible benchmarks, framework-driven application evals, and live-traffic observability**. The landscape now centers on roughly a dozen serious open-source projects, with **RAGAS, DeepEval, lm-evaluation-harness, Promptfoo, and Inspect AI** representing the most active poles of activity, while **Langfuse, Phoenix, and Opik** lead the open-source observability tier. Choosing correctly matters: a wrong-fit framework wastes engineering quarters, while the right one becomes a CI gate that catches regressions before customers do. This guide synthesizes 17+ frameworks, 50+ metrics, and the use-case mappings that connect them — based on research current to May 2026.

1. What LLM evaluation is and why it matters

LLM evaluation is the systematic measurement of language-model outputs against quality, safety, and operational criteria. Unlike classical ML evaluation — where a held-out test set and an F1 score usually suffice — LLM outputs are **open-ended, stochastic, multi-turn, and often reference-free**, which breaks most traditional metrics. The field has responded with three innovations: **LLM-as-a-judge** (using a strong model to grade another model's output), **decomposition into atomic claims** (so "faithfulness" can be measured per fact), and **trace-level evaluation** (scoring multi-step agent trajectories rather than just final answers).

Why it matters now: production LLM apps fail in ways unit tests can't catch — hallucinations, prompt-injection jailbreaks, silent retrieval drift, multi-turn knowledge loss, runaway agent costs. IBM's 2025 report found 13% of organizations had AI-related security breaches and 97% lacked proper AI access controls. Meanwhile benchmark contamination has rendered MMLU and HumanEval scores unreliable for top models, forcing the field toward private golden datasets, dynamic benchmarks (Chatbot Arena, LiveBench), and rigorous test-augmented variants (HumanEval+, MMLU-Pro). (Galileo AI)

The 2026 landscape splits into four tiers:



2. The frameworks, in depth

The 17+ frameworks span academic benchmarks (designed for reproducible model leaderboards) and application frameworks (designed for shipping LLM products). The distinction matters: **lm-evaluation-harness** can tell you GPT-5 scores 87.3% on MMLU-Pro but

cannot tell you your customer-support chatbot has a knowledge-retention problem. The table below summarizes everything; deeper notes follow.

Framework	License	Stars (May 2026)	Primary purpose	Owner
RAGAS	Apache-2.0	~13.9k	Reference-free RAG evaluation	Exploding Gradients / VibrantLabs
DeepEval	Apache-2.0	~13.6k	Pytest-style LLM unit testing	Confident AI (YC)
TruLens	MIT	~3.1k	Feedback functions, RAG Triad	Snowflake
Giskard	Apache-2.0	~5.3k	Safety, red-team, EU compliance	Giskard SAS (Paris)
UpTrain	Apache-2.0	~2.3k	Local-first dashboard + RCA	UpTrain (YC)
lm-evaluation-harness	MIT	~12.2k	Benchmark suite for base models	EleutherAI
OpenAI Evals	MIT	~17.5k	Registry of evals with model-graded support	OpenAI
OpenAI simple-evals	MIT	~3-4k	Reference impls of OpenAI's reported scores	OpenAI (unmaintained)
HELM	Apache-2.0	~2.6k	Holistic 7-axis evaluation	Stanford CRFM
BIG-Bench	Apache-2.0	~3.1k	200+ task capability profiling	Google + community
MT-Bench	Apache-2.0	(in FastChat ~39k)	Multi-turn LLM-as-judge benchmark	LMSYS / LMArena
AgentBench	Apache-2.0	several thousand	Multi-environment agent eval	Tsinghua KEG
EvalPlus	Apache-2.0	~1.7k	Rigorous code eval (HumanEval+ / MBPP+)	EvalPlus team
HumanEval	MIT	~3k	Canonical 164-problem code benchmark	OpenAI

Framework	License	Stars (May 2026)	Primary purpose	Owner
Promptfoo	MIT	~21k	YAML CI prompt + red-team testing	Promptfoo (acquired by OpenAI Mar 2026)
MLflow LLM Evaluate	Apache-2.0	~20k+ (mlflow)	LLM evals inside MLflow tracking	Databricks
Arize Phoenix	Elastic-2.0 / Apache-2.0 evals	~9k+	OTel-native observability + evals	Arize AI
W&B Weave	Apache-2.0 (SDK)	~1k	Tracing + evals in W&B	Weights & Biases
Inspect AI	MIT	~1.3k (rapid growth)	Frontier safety / agent evals	UK AI Security Institute
Langfuse	MIT (core)	~24.6k	Open-source LangSmith alternative	Langfuse (acquired by ClickHouse Jan 2026)
Opik	Apache-2.0	~12.5k+	OSS observability + evals	Comet ML
Braintrust	Proprietary (Autoevals OSS MIT)	n/a (closed)	Evaluation-first SaaS	Braintrust Data
LangSmith	Closed-source	n/a (openevals ~1k)	LangChain SaaS evals + tracing	LangChain Inc
Confident AI	Closed (DeepEval = OSS)	(DeepEval 13.6k)	Hosted DeepEval cloud	Confident AI (YC)

RAG-focused frameworks

RAGAS (github.com/explodinggradients/ragas) is the de facto standard for **reference-free RAG evaluation**. Introduced in a 2023 paper (arXiv 2309.15217, EACL 2024), it pioneered the pattern of *decomposing answers into atomic claims and verifying each against retrieved context*. **Strengths:** it's lightweight, integrates with virtually every observability tool (Langfuse, Phoenix, LangSmith, Helicone), and ships a synthetic test-set generator that builds RAG questions from your knowledge base. **Weaknesses:** heavy LLM-as-judge cost at scale, no first-party UI, and the late-2025 rebrand to "vibrantlabs" caused doc confusion. Best when your sole concern is RAG quality and you live in notebooks or pytest. ([github](#) +2)

DeepEval (github.com/confident-ai/deepeval) describes itself as "**Pytest for LLMs.**" With 50+ built-in metrics — RAG, agentic, multi-turn, multimodal, safety — and 8,800+ commits across 236+ contributors, it is the broadest application-eval framework in 2026. The `@observe` decorator does component-level eval on agent traces; **DAG metrics** give deterministic conditional scoring; **DeepTeam** is the sister red-team library (40+ vulnerabilities, OWASP-mapped). Pairs optionally with **Confident AI** for hosted dashboards. Best for teams wanting CI/CD-driven LLM regression testing with one framework covering everything. (GitHub + 9)

TruLens (github.com/truera/trulens) — now Snowflake-maintained after the 2024 TruEra acquisition — pioneered the "**RAG Triad**" (context relevance + groundedness + answer relevance) that almost every other tool now mirrors. MIT-licensed (most permissive in this list), OpenTelemetry-native, with feedback functions attached to spans. Smaller community (~3.1k stars), but the cleanest semantic decomposition of RAG failure modes. Best when you specifically want triad framing or are in the Snowflake/Cortex ecosystem. (Atlan + 4)

Giskard (github.com/Giskard-AI/giskard) is the **EU-compliance and red-team specialist**. Founded in Paris, EU-funded (Bpifrance + EIC Accelerator), with audit-ready artifacts for the EU AI Act and OWASP LLM Top-10. Its **LLM Scan** auto-detects 40+ vulnerabilities (jailbreaks, prompt injection, PII leakage, stereotypes), and **RAGET** is one of the best dedicated synthetic test-set generators. The v3 rewrite (April 2026) is alpha but cleaner. Trusted by AXA, BNP Paribas, Michelin. Best for regulated industries. (Giskard + 6)

UpTrain (github.com/uptrain-ai/uptrain) offers a **fully local self-hosted dashboard** with strong root-cause analysis (RCA) on failure patterns — rare among OSS competitors. With ~2.3k stars and slowed cadence in 2025–2026 (founders have spun up CombineHealth), it's the smallest in this tier but valuable for teams that need privacy and 20+ checks out-of-the-box. (github + 3)

Benchmark and research frameworks

EleutherAI's lm-evaluation-harness (github.com/EleutherAI/lm-evaluation-harness) is the **academic gold standard** — the engine behind Hugging Face's Open LLM Leaderboard. ~12.2k stars, MIT, 200+ tasks via YAML configs, supports HuggingFace, vLLM, OpenAI-compatible APIs, NeMo, MLX backends. Every task is versioned for reproducibility. Used by NVIDIA, Cohere, BigScience, Mosaic ML, and cited in hundreds of papers. The right tool for pre-training and fine-tuning leaderboards but **not for production app evaluation**. (GitHub + 6)

HELM (github.com/stanford-crfm/helm) goes further than accuracy. It defines a three-axis taxonomy (Scenarios × Adaptation × Metrics) with **seven evaluation dimensions**: accuracy, calibration, robustness, fairness, bias, toxicity, and efficiency. Sub-benchmarks include **VHELM** (vision), **HEIM** (text-to-image), **MedHELM**, **AIR-Bench 2024** (safety/regulation). It's heavyweight but the most comprehensive multi-dimensional view of a model. (GitHub)

BIG-Bench (github.com/google/BIG-bench) — 214 tasks from 442+ authors across 132 institutions — is now mostly accessed via its **BBH** (BIG-Bench Hard, 23 hardest tasks) and **BBEH** (Extra Hard, 2025 successor) subsets, since the full benchmark is saturated by frontier models. BBH is bundled into both lm-evaluation-harness and the HF Open LLM Leaderboard v2. (arxiv)

MT-Bench (in [lm-sys/FastChat](#)) introduced the modern **LLM-as-judge** methodology in NeurIPS 2023: 80 multi-turn prompts × 8 categories, scored by GPT-4 in single, pairwise, or reference-guided mode. ~85% agreement with human experts. Now superseded for separability by **Arena-Hard-Auto** (LMArena, ~98.6% Chatbot-Arena correlation, 3× more separable) and the gold-standard **Chatbot Arena** itself, which uses Bradley-Terry scoring with Style Control to debias length/markdown effects. (GitHub +2) (Grokipedia)

AgentBench ([github.com/THUDM/AgentBench](#)) — ICLR 2024 — pioneered LLM-as-Agent benchmarking across 8 environments (Linux shell, SQL, knowledge graph, card game, lateral thinking, ALFWorld, WebShop, Mind2Web). For agentic SOTA in 2026, however, the field has shifted to **SWE-bench Verified** (real GitHub issues; 87–94% on top models), **GAIA / GAIA2** (general-assistant tool use; humans 92% vs. early GPT-4 15%), and **τ-bench / WebArena / OSWorld**. (LLM Leaderboard)

EvalPlus ([github.com/evalplus/evalplus](#)) and **HumanEval** ([github.com/openai/human-eval](#)) are the canonical code-generation evaluators. EvalPlus augments HumanEval with 80× more tests and MBPP with 35× more, dropping pass@1 by 19–29% on average — **revealing brittle code that base HumanEval misses**. EvalPerf adds efficiency benchmarking. EvalPlus is the standard for serious code-LLM papers in 2026. (GitHub +2 + 3)

Inspect AI ([github.com/UKGovernmentBEIS/inspect_ai](#)) — released May 2024 by the UK AI Security Institute, led by RStudio founder JJ Allaire — has rapidly become the **frontier-AI safety standard**. Used by Anthropic, DeepMind, xAI, METR, Apollo Research. Its Dataset → Solver → Scorer abstraction, sandboxed Docker execution, Agent Bridge for OpenAI/LangChain/Pydantic AI, and 200+ pre-built evals (([inspect_evals](#)) registry) make it the go-to for sandboxed agent evaluation. ~1.3k stars but among the highest-velocity repos in the safety segment. (GitHub + 7)

Production observability + eval platforms

Langfuse ([github.com/langfuse/langfuse](#)) is the **leading open-source LangSmith alternative** with ~24.6k stars (the largest in this category). MIT-licensed, free unlimited self-host, OpenTelemetry-based. Acquired by ClickHouse in January 2026 as part of their \$400M Series D, but the project remains open. Tracing + evals + prompt management + datasets + experiments all unified. (Doolpa + 5)

Arize Phoenix ([github.com/Arize-ai/phoenix](#)) is the **OpenTelemetry-native** option, ~9k+ stars, Elastic License v2 (source-available; Apache-2.0 for the evals package). Auto-instruments LangChain, LlamaIndex, OpenAI Agents SDK, Claude Agent SDK, LangGraph, CrewAI, DSPy. Excellent notebook UX; 2.5M+ monthly downloads. The right pick when OpenTelemetry exportability and framework-agnosticism matter. (GitHub +2 + 2)

Opik ([github.com/comet-ml/opik](#)) launched September 2024, hit 12.5k+ stars within a year, and is the fastest-growing OSS LangSmith alternative — Apache-2.0, with PyTest CI integration, agent optimizer, and "Ollie" coding agent for trace-driven fixes. (Trilogyequity + 2)

LangSmith is the closed-source LangChain SaaS — frictionless if you're already on LangChain/LangGraph but vendor-locked. **Braintrust** is the closed-source "evaluation-first"

platform used by Notion, Stripe, Vercel, Airtable; its **Autoevals** library is MIT-licensed OSS. **W&B Weave** (~1k SDK stars but huge W&B install base) targets teams already using Weights & Biases for experiment tracking. **Confident AI** is the hosted layer atop DeepEval — SOC 2 Type II + HIPAA + on-prem. [Braintrust](#) [Confident AI](#)

MLflow LLM Evaluate (github.com/mlflow/mlflow) shines for **hybrid ML+LLM teams already invested in MLflow**. The new `mlflow.genai.evaluate()` API (MLflow 3, with `@scorer` decorator) integrates with MLflow Tracing and Review App. Free, fully OSS, Databricks-supported. [MLflow](#) [Databricks](#)

Promptfoo (github.com/promptfoo/promptfoo) — ~21k stars, MIT, **acquired by OpenAI in March 2026** — is the YAML/CLI-driven prompt testing tool with the strongest CI/CD and red-teaming story (100+ attack plugins mapped to OWASP LLM Top-10). Used by 300,000+ developers and 127 Fortune 500 customers. [GitHub](#) [AppSec Santa](#)

3. The metrics catalog

Metrics divide into seven families. Each family solves a different evaluation problem; serious LLM evaluation combines metrics across families rather than relying on any single number.

graph LR

M[LLM Eval Metrics] --> R[RAG]
M --> H[Hallucination & Truth]
M --> S[Safety & Alignment]
M --> N[Reference-based NLG]
M --> J[LLM-as-Judge]
M --> P[Probabilistic]
M --> C[Code & Agent]
M --> O[Operational]

R --> R1[Faithfulness]
R --> R2[Answer Relevancy]
R --> R3[Context Precision]
R --> R4[Context Recall]
R --> R5[Context Relevancy]
R --> R6[Noise Sensitivity]

H --> H1[Hallucination metric]
H --> H2[TruthfulQA MC1/MC2]
H --> H3[SelfCheckGPT]

S --> S1[Toxicity]
S --> S2[Bias]
S --> S3[PII Leakage]
S --> S4[Jailbreak ASR]

N --> N1[BLEU]
N --> N2[ROUGE]
N --> N3[BERTScore]
N --> N4[METEOR / chrF]
N --> N5[Exact Match / F1]
N --> N6[MoverScore]
N --> N7[COMET]

J --> J1[G-Eval]
J --> J2[QAG Score]
J --> J3[DAG Metric]
J --> J4[Prometheus]
J --> J5[MT-Bench judge]

P --> P1[Perplexity]
P --> P2[Log-likelihood]

C --> C1[Pass@k]
C --> C2[HumanEval+/MBPP+]
C --> C3[Tool Correctness]
C --> C4[Task Completion]
C --> C5[Trajectory Eval]
C --> C6[Step Efficiency]

O --> O1[Latency p50/p95/p99]

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0 --> 02[Cost per query]
0 --> 03[Throughput]
```

RAG metrics

Faithfulness (Ragas, DeepEval, TruLens groundedness) measures whether response claims are inferable from retrieved context. The algorithm decomposes the response into atomic claims, then asks an LLM-judge whether each claim is supported. **Score = supported claims / total claims**, range 0-1. It's reference-free, directly targets hallucinations relative to sources, and is per-claim explainable — but a faithful-but-incomplete answer can still score 1.0, and the metric doesn't catch wrong-but-cited facts. (Redis + 4)

Answer Relevancy measures intent alignment between the user's question and the response. **Ragas** uses reverse-question generation: the LLM creates N artificial questions from the response, embeds both, and computes mean cosine similarity. **DeepEval** instead extracts statements from the answer and classifies each as relevant/irrelevant to the input. Reference-free; doesn't measure truthfulness. (ragas)

Context Precision evaluates retriever ranking quality — are the most relevant chunks at the top? Calculated as a weighted mean of Precision@k across retrieved positions. Tunes rerankers and top-K. **Context Recall** measures retrieval completeness against a reference: did all needed evidence get surfaced? Tunes embedding model and chunking strategy. **Context Relevancy** is the reference-free version (per-chunk relevance to query). **Context Entities Recall** is entity-focused (% of named entities in reference present in retrieved contexts) — useful for legal/medical/finance. (ragas)

Noise Sensitivity (Ragas, from Amazon's RAGChecker) measures how often the system makes wrong claims when given relevant or irrelevant contexts. **Lower is better**. Stress-tests RAG against distractors. (ragas)

RAGAS Score is the harmonic-mean composite of faithfulness + answer relevancy + context precision + context recall. Useful for dashboards but hides which component is failing. (Medium)

Hallucination and truthfulness

DeepEval's Hallucination metric computes the fraction of provided context documents that contradict the output (lower is better, threshold-as-maximum). Different from RAG faithfulness — this targets the LLM itself against a known-true reference set. (Deepeval)

TruthfulQA (817 adversarially crafted misconceptions across 38 categories) evaluates resistance to imitating human falsehoods. Modes: generation (GPT-judge scores Truth and Truth×Info), MC1 (single correct), MC2 (probability mass on true answers — most-cited), MC3 (true ranked above false). Saturated for top models but still standard. (ACL Anthology + 5)

SelfCheckGPT is the **black-box hallucination detector** — generate the main response plus N stochastic samples, then score consistency (via BERTScore, NLI entailment, n-gram, QA, or LLM-prompt variants). Premise: known facts are self-consistent across samples; hallucinations

diverge. Higher score = more likely hallucinated. Works on closed APIs without ground truth, at the cost of $N \times$ generation per query. (OpenReview + 3)

Safety and alignment

Toxicity is measured by either LLM-judge (DeepEval extracts opinions and classifies each), classifier (Detoxify multilingual transformers, Perspective API CNN/UTC), or benchmark (RealToxicityPrompts, ToxicChat). All have known biases against identity terms; LLM judges are subjective. Use as a maximum threshold (default 0.5). (Sebastianduerr + 4)

Bias detection covers gender, racial, political, religious, geographic. Implementations: DeepEval LLM-judge opinion extraction, **StereoSet/CrowS-Pairs** (probability of stereotypical vs anti-stereotypical preference), **BBQ** (Bias Benchmark for QA, accuracy in ambiguous vs disambiguated settings).

PII Leakage (DeepEval (PIILeakageMetric)) extracts PII from output and classifies real vs placeholder. Critical for GDPR/HIPAA/CCPA. **Jailbreak Resistance** is measured as **Attack Success Rate (ASR)** = successful jailbreaks / total attempts, evaluated via classifier (LlamaGuard, ShieldGemma) on output. Attack suites: HarmBench, JailbreakBench, AdvBench, plus dynamic methods (GCG, AutoDAN, PAIR, Crescendo). (deepeval)

Reference-based NLG metrics

These are the classical metrics — cheap, deterministic, but increasingly weak for LLM-style generation.

BLEU (Papineni 2002) computes clipped n-gram precision with brevity penalty: $BP \times \exp(\sum w_n \log p_n)$, $N=4$, weights $1/N$. Range 0-1 (reported 0-100). Standard for MT but has poor correlation with humans for paraphrastic LLM outputs. (GeeksforGeeks) (ACL Anthology)

ROUGE is recall-oriented for summarization. **ROUGE-1/2** are unigram/bigram overlap; **ROUGE-L** uses Longest Common Subsequence (capturing structure without contiguity). Fast, standard for CNN/DailyMail and XSum benchmarks; surface-level and misses paraphrasing. (Medium)

BERTScore (Zhang 2020) computes token-level cosine similarity using contextual BERT embeddings, with greedy matching and optional IDF weighting. P = mean(max-similarity for each candidate token); R symmetric; $F = 2PR/(P+R)$. Captures synonyms; high human correlation; GPU-expensive. (AWS)

METEOR combines unigram precision + recall + WordNet synonym matching + word-order penalty: $METEOR = F_mean \times (1 - \gamma(\text{chunks/matches})^\alpha)$, with $\alpha=0.9$ recall-weighted. Better human correlation than BLEU at sentence level; English-centric. (Vinija) (Vinija)

Exact Match and **token-level F1** are SQuAD standards — EM is brittle; F1 forgiving but bag-of-tokens.

MoverScore uses Earth Mover's Distance over BERT embeddings: $1 - WMD_n(x, \hat{x})$, with IDF weights. Soft alignment beats BERTScore's hard greedy match for dialogue and image captioning. (Vinija) (arXiv)

COMET is the modern MT metric — neural model fine-tuned on human DA judgments. Best human correlation in WMT22, but blind to numeric/named-entity errors. **Pair with chrF + BLEU** for diagnostic backup.

LLM-as-judge metrics

G-Eval (Liu 2023, DeepEval implementation) is the most adopted LLM-judge metric in 2026 — DeepEval reports ~10M G-Eval evaluations per month. Algorithm: (1) LLM auto-generates evaluation steps from a natural-language criteria string, (2) prompts the judge for an integer score 1-5 or 1-10, (3) **token-probability normalization**: weighted sum of output-token probabilities $\times$ score values, reducing discretization bias. Original paper showed Pearson > 0.5 with humans on summarization vs ~ 0.2 for BLEU/ROUGE. (deepeval + 4)

QAG Score (Question-Answer-Generation) is the underlying technique behind DeepEval's faithfulness, answer relevancy, hallucination, bias, toxicity, and summarization metrics. Decomposes evaluation into close-ended yes/no questions, then aggregates positive answers / total. Constrained outputs reduce stochasticity vs free-form LLM scoring.

DAG (Deep Acyclic Graph) metric in DeepEval lets you build a deterministic decision tree of LLM judgments — TaskNodes (extraction), BinaryJudgementNodes, NonBinaryJudgementNodes, VerdictNodes (leaf scores or delegate to G-Eval). Topological traversal yields a reproducible score. Best for production CI gates needing multi-criteria deterministic scoring. (deepeval + 2)

Prometheus 1 / 2 / 2-BGB are open-source judge LMs (Llama-2 13B $\rightarrow$ Mistral 7B/8x7B). Direct Assessment scores 1-5 with rubric + reference; Pairwise Ranking picks A or B. Prometheus 2-BGB surpasses Claude-3-Opus on absolute grading; Pearson 0.897 with humans (vs GPT-4's 0.882) on 45 custom rubrics. The standard for cost-controlled, reproducible academic LLM-judging. (arXiv)

MT-Bench LLM-Judge uses GPT-4 in single/pairwise/reference-guided modes on 80 multi-turn prompts. 85% agreement with human experts. Known biases: position, verbosity, self-enhancement, weak math reasoning.

Probabilistic metrics

Perplexity = $\exp(-1/t \cdot \sum \log p(x_i | x_{<i}))$, the exponentiated cross-entropy. ≥ 1 , lower is better. Use sliding-window for long contexts. Tokenizer-dependent (cannot compare across tokenizers); doesn't measure correctness. Standard for pre-training evaluation. (Hugging Face) (Hugging Face)

Log-likelihood is the foundation of multi-choice benchmarks (MMLU, HellaSwag, ARC) — $\log p(y_{\text{correct}}) - \log p(y_{\text{wrong}})$. Requires logit access (not available on closed APIs).

Code generation metrics

Pass@k (Chen 2021, OpenAI Codex paper) is the unbiased estimator: $1 - C(n-c, k)/C(n, k)$, where n samples are generated and c pass. Original HumanEval used $n=200$ with temperatures

0.2/0.6/0.8 for k=1/10/100. Functional correctness via execution beats text similarity for code.

(DataCamp + 2)

HumanEval pass rate aggregates pass@k over 164 hand-written Python problems with ~7.7 unit tests each — saturated (>90% Pass@1) on frontier models in 2026, motivating EvalPlus's 80× test augmentation. (Medium + 2)

EvalPlus / HumanEval+ / MBPP+ drops top-model Pass@k by 19–29% on average and changes model rankings (e.g., WizardCoder-CodeLlama beats ChatGPT post-EvalPlus despite tying pre-EvalPlus). **EvalPerf** introduces **eff@k** for execution efficiency. (arxiv + 2)

Agent metrics

Tool Correctness (DeepEval) is mostly deterministic: compare `tools_called` vs `expected_tools` with configurable strictness — name match, name+arguments, name+arguments+output, with or without ordering. Optional LLM-judge layer scores optimality vs available alternatives. **Ragas Tool Call Accuracy** and **Tool Call F1** are set-based variants.

(deepeval + 2)

Task Completion (DeepEval, also called Goal Accuracy in Ragas) operates on the full agent trace — extract task from input, examine all tool calls, and have an LLM-judge produce a 0–1 outcome score. (deepeval) (Confident AI)

Trajectory Evaluation decomposes agent quality into Plan Quality (logical, scoped, dependency-aware), Plan Adherence (did the agent follow its plan?), and Trajectory Match (actual vs reference action sequence). Implementations: DeepEval, LangSmith trajectory eval, Galileo, Arize. (DeepEval) (DeepEval)

Step Efficiency (DeepEval) compares actual step count and quality against minimum needed — surfaces wasteful loops and redundant tool calls. Critical for cost control in production agents.

Conversational metrics (DeepEval)

Conversation Completeness = fulfilled user intentions / total user intentions across the session — the most important multi-turn metric. **Knowledge Retention** = assistant turns without forgetting / total assistant turns. **Role Adherence** = turns staying in persona / total. **Topic Adherence** = (TP + TN) / total QA pairs (rewards correct refusals on off-topic queries). **Turn Relevancy** = relevant turns / total in sliding window. (deepeval + 7)

Operational metrics

Latency is reported as percentiles (p50, p95, p99), with **Time-to-First-Token (TTFT)** and **Time-Per-Output-Token (TPOT)** for streaming. **Cost per query** = (prompt_tokens · c_in) + (completion_tokens · c_out); **quality-adjusted cost** = total cost / # successful tasks. **Throughput** in tokens/second. Captured by every observability platform; enforce as SLA gates.

Summarization-specific metrics

DeepEval's Summarization Score = min(Alignment, Coverage), where Alignment uses QAG to

verify summary claims against the document, and Coverage uses QAG to verify the document's key facts are answerable from the summary. Reference-free; combines faithfulness and completeness.

G-Eval on SummEval dimensions — Coherence, Consistency, Fluency, Relevance — gives Spearman 0.51-0.59 with humans vs <0.20 for ROUGE/BERTScore. (Openai)

Classification metrics

Standard ML metrics still apply: Accuracy, Precision, Recall, F1 (per-class plus macro/weighted), AUC-ROC, AUC-PR, MCC, Cohen's κ . **When LLM-judge helps for classification:** open-label tasks, subjective categories, multilingual coverage without retraining. **When NOT to use it:** when fine-tuned encoder models (DeBERTa, ModernBERT) are available — they're 10-100× cheaper and often more accurate.

4. Use case → metric → framework mapping

This is the most actionable part of the guide. Each use case names the metrics that matter, the frameworks that excel, and the workflow pattern.

flowchart TB

UC{What are you
evaluating??}

UC -->|RAG| RAG[Faithfulness + Answer Relevancy
+ Context Precision + Context

UC -->|Chatbot| CHAT[Conversation Completeness
+ Knowledge Retention
+ R

UC -->|Code Gen| CODE[Pass@k on HumanEval+MBPP+
+ BigCodeBench + SWE-bench]

UC -->|Summarization| SUM[Summarization Score
+ G-Eval Coherence/Consistency

UC -->|Classification| CLS[Accuracy/Precision/Recall/F1
+ Confusion Matrix]

UC -->|Agents| AG[Task Completion
+ Tool Correctness
+ Trajectory + Step

UC -->|Safety| SAF[Toxicity + Bias + PII
+ Jailbreak ASR]

UC -->|General Bench| GB[MMLU-Pro + GPQA + IFEval
+ BBH + MATH + MUSR]

UC -->|QA| QA[EM + F1 + SAS
or LLM-judge correctness]

UC -->|Translation| MT[COMET primary
+ chrF + BLEU diagnostic]

RAG -->|frameworks| RAGFW[RAGAS / DeepEval / TruLens / Phoenix]

CHAT -->|frameworks| CHATFW[DeepEval Conversational / LangSmith / Langfuse]

CODE -->|frameworks| CODEFW[EvalPlus / BigCodeBench / SWE-bench / lm-eval-harnes

SUM -->|frameworks| SUMFW[DeepEval / Ragas / OpenAI Evals]

CLS -->|frameworks| CLSFW[scikit-learn / HF evaluate / Evidently]

AG -->|frameworks| AGFW[DeepEval / LangSmith / Phoenix / Inspect AI]

SAF -->|frameworks| SAFFW[Giskard / DeepTeam / Garak / PyRIT / Promptfoo]

GB -->|frameworks| GBFW[lm-evaluation-harness / HELM / Inspect AI]

QA -->|frameworks| QAFW[HF evaluate squad / DeepEval / Ragas]

MT -->|frameworks| MTFW[unbabel-comet / sacrebleu / HF evaluate]

RAG systems

Metrics that matter: Faithfulness (catch hallucinations), Answer Relevancy (catch off-topic),

Context Precision (retriever ranking), Context Recall (retriever completeness), Context Relevancy (online reference-free), Factual Correctness (vs ground truth when available). The arXiv telecom RAG study (2407.12873) found Faithfulness + Factual Correctness *together* are the most reliable — either alone misleads. [Confident AI](#) [arxiv](#)

Recommended frameworks: RAGAS for notebook iteration, DeepEval for CI/CD, TruLens for production tracing, Phoenix for OTel-native observability.

Workflow: Build a 50–200 question golden set from real logs (or via Ragas synthetic generation). In CI, run DeepEval thresholds (faithfulness ≥ 0.8 , answer relevancy ≥ 0.85). Tune retriever using context precision/recall; tune prompt using faithfulness + answer relevancy. In production, sample-evaluate live traffic with Phoenix or Langfuse + Ragas faithfulness.

Key pitfall: No framework distinguishes a *factually wrong* retrieved context from a correct one — garbage-in still produces high faithfulness. Pair with knowledge-base freshness checks. [Atlan](#)

Chatbots and conversational AI

Metrics that matter: Conversation Completeness (most important — if the task isn't done, nothing else matters), Knowledge Retention (does the bot remember what the user said?), Role Adherence (stays in persona), Topic Adherence (stays in domain), Turn Relevancy, Conversational G-Eval for custom criteria. [Confident AI](#) [GitHub](#)

Recommended frameworks: DeepEval has native [ConversationalTestCase](#) and [ConversationalGolden](#) with all of these out-of-the-box plus a multi-turn simulator. LangSmith is best for LangGraph chat agents (session tracing, thread-level evals). Langfuse is the OSS alternative for multi-turn conversation tracking with annotation queues. [deepeval](#) [Deepeval](#)

Workflow: Define [ConversationalGoldens](#) for scenarios (e.g., "angry user requesting refund"), simulate N variations, gate releases on aggregate scores. [Deepeval](#)

Code generation

Metrics that matter: Pass@1 and Pass@10 (functional correctness), efficiency (eff@k from EvalPerf when relevant), style/maintainability (LLM-judge or linter — only when needed; lead with execution-based metrics).

Benchmarks by use case:

Benchmark	Scope	When to use
HumanEval+ (EvalPlus)	164 Python problems with 80× tests	Standard rigor; replaces vanilla HumanEval
MBPP+ (EvalPlus)	378 entry-level Python with 35× tests (GitHub)	Pair with HumanEval+ for breadth
BigCodeBench	Library-aware function calls	Realistic / less saturated
LiveCodeBench	Time-stamped contest problems	Anti-contamination (post-cutoff slice)
SWE-bench Verified	500 real GitHub issues, human-validated (GitHub)	Production-style coding agents
HumanEval-X / MultiPL-E	Multi-language ports	Beyond Python
HumanEval Pro / MBPP Pro	Self-invoking variants	Tests reasoning beyond memorization

Recommended frameworks: EvalPlus is the de-facto execution harness; BigCodeBench harness (Docker), bigcode-evaluation-harness, lm-evaluation-harness code tasks, SWE-bench runner.

Summarization

Metrics that matter: Lead decisions on G-Eval scoring of SummEval's four dimensions (Coherence, Consistency, Fluency, Relevance) plus LLM faithfulness. Report ROUGE and BERTScore for backwards comparability only. Watch for G-Eval's known bias toward longer GPT-style outputs. (Openai)

Recommended frameworks: DeepEval (SummarizationMetric, GEval), Ragas faithfulness, OpenAI Evals cookbook recipes, HF evaluate for ROUGE/BERTScore.

Classification

Metrics that matter: Accuracy (balanced data only), per-class Precision/Recall/F1 (imbalanced), Confusion Matrix for diagnosis, AUC-ROC / PR-AUC for probability outputs, Cohen's κ for human-label agreement.

When LLM-judge helps: open-label tasks, subjective categories, multilingual coverage, label bootstrapping. **When not:** any task with adequate labeled data and a fine-tuned encoder available.

Recommended frameworks: scikit-learn, Hugging Face evaluate, Evidently AI for drift + classification monitoring, DeepEval GEval for custom criteria.

Agentic / multi-step tasks

Metrics that matter — both outcome AND trajectory:

Type	Metrics
Outcome (final state)	Task success rate, exact answer match, end-state correctness
Trajectory (process)	Tool selection accuracy, argument correctness, dependency/order, step-level effectiveness, unnecessary-step count, cost/latency

AgentBench (Liu 2024) showed final-state alone is insufficient: a "right answer" can mask unsafe or invalid actions. [arXiv](#)

Benchmarks: GAIA / GAIA2, AgentBench, SWE-bench Verified, WebArena / VisualWebArena, MINT, τ -Bench, TRAJECT-Bench, TRAIL, AgentProcessBench.

Recommended frameworks: DeepEval ([ToolCorrectnessMetric](#)), [TaskCompletionMetric](#), [StepEfficiencyMetric](#)), LangSmith trajectory eval / AgentEvals, Arize Phoenix (deepest agent trace capability), Inspect AI for sandboxed safety-critical agent eval. [deepeval](#) [TECHSY](#)

Workflow: Define ground-truth trajectories for 50 scenarios. Score each run on (a) goal completion (LLM judge), (b) tool selection precision/recall vs expected, (c) argument correctness via schema validation, (d) trajectory length penalty.

Safety and alignment

Risk dimensions: Toxicity, Bias (gender/race/religion), PII leakage, Jailbreak resistance, Prompt injection / data exfiltration, Refusal correctness (refuse harmful + answer benign — over-refusal is a real risk).

Benchmarks: ALERT (45k red-team prompts), HarmBench, AdvBench, ToxicChat, TruthfulQA, HELM Safety scenarios, DecodingTrust. [arxiv](#)

Tool comparison (arXiv 2410.16527):

Tool	Strength	Weakness
Garak (NVIDIA)	37+ probe modules; static + dynamic	Less multi-turn
Promptfoo	50+ vuln types; CI/CD-native; YAML	Less advanced attacks
PyRIT (Microsoft)	Multi-turn (Crescendo), multimodal	High prompt-engineering effort
Giskard	Easy LLM attack generation; guardrails	Locked to GPT for attacker
DeepTeam	40+ vuln types, OWASP-mapped, integrates with DeepEval	Younger ecosystem
CyberSecEval (Meta)	Cybersec-specific	Narrow scope

Best practice: Run Garak/Promptfoo for static scan + PyRIT for multi-turn + custom suite for app-specific risks before every model upgrade. Add a runtime guardrail layer (NeMo Guardrails, Llama-Guard, Galileo Luna, Lakera Guard).

General benchmarking

Hugging Face Open LLM Leaderboard v2 (current standard, replaced v1 in 2024):

Benchmark	Tests
IFEval	Strict instruction-following
BBH	23 challenging reasoning subtasks
MATH (lvl-5)	Competition math
GPQA	Graduate-level science (Google-proof)
MUSR	Multi-step soft reasoning
MMLU-Pro	Harder, expert-reviewed MMLU

V1 (MMLU, ARC, HellaSwag, TruthfulQA, Winogrande, GSM8K) is **saturated and likely contaminated** — useful for legacy comparison only.

Standard runner: **lm-evaluation-harness** is the reference. **HELM** is the alternative for multi-axis (accuracy + safety + robustness + fairness + bias + toxicity + efficiency). **OpenAI simple-evals** for benchmark replication only.

Question answering

Short / extractive QA: Exact Match + token-level F1 (SQuAD standards). Both fail at semantic equivalence ("the French capital" vs "Paris").

Open-domain / long-form QA: Add **Semantic Answer Similarity** (deepset cross-encoder, EMNLP 2021), **BERTScore**, or **LLM-judge correctness**. For SQuAD2.0 also track abstention rate on unanswerable questions. ([GitHub](#))

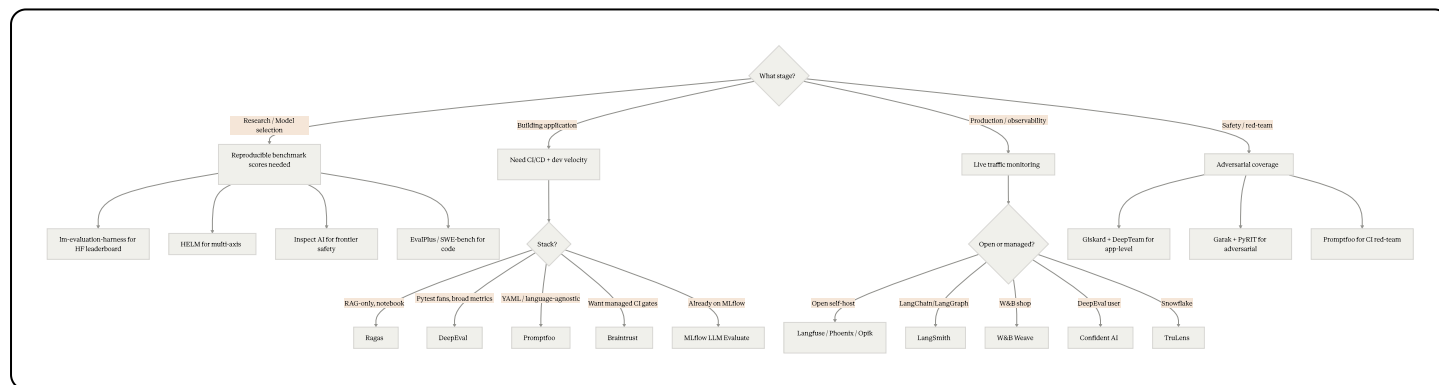
Translation

Lexical: BLEU, chrF/chrF++ (better for morphologically rich languages), TER. **Neural state-of-the-art:** **COMET** (and reference-free COMET-Kiwi, xCOMET) — best human correlation in WMT22 but blind to numeric/named-entity errors. **BLEURT** is the BERT alternative.

Best practice: COMET as primary, complement with chrF + BLEU diagnostic. For literary/legal/medical translation, add LLM-judge with rubric.

5. Choosing the right framework

The single most useful question: *what stage are you in?*



The decision criteria checklist:

Criterion	Question	Implication
Open vs managed	Need data residency / no third-party?	Open: Langfuse, Phoenix, DeepEval, Opik. Managed: LangSmith, Braintrust, Confident AI, Galileo.
CI/CD integration	Block PRs on regression?	Pytest-native: DeepEval. Built-in gates: Braintrust. YAML/CLI: Promptfoo.
Framework lock-in	LangChain/LangGraph user?	LangSmith zero-config. Otherwise prefer OTel-native (Phoenix, Langfuse).
Metric breadth	RAG + agent + multi-turn + safety in one?	DeepEval (50+); Ragas is RAG-only.
Multi-turn / agent depth	Multi-step tool use?	DeepEval conversational + Phoenix / LangSmith trajectory.
Cost model	Predictable spend at scale?	Trace-billed SaaS penalizes agents (50 spans = 50× cost). Pydantic Self-host is cheapest at scale.
Language / SDK	Python only? TypeScript?	Most are Python-first; LangSmith, Braintrust, Langfuse have strong TS SDKs.
Learning curve	Engineers vs PMs writing evals?	DeepEval has steep curve. Braintrust/Maxim aim at non-engineers.
Red-team needs	OWASP coverage?	DeepTeam, Garak, PyRIT, Promptfoo, Giskard.

When to use what kind of evaluation:

- **Public benchmarks** (MMLU-Pro, HumanEval+, GPQA): foundation-model selection, model-card reporting. *Never* as the sole production gate.
- **Custom golden datasets** (50–500 task-specific examples): the single most important production eval. Build from real user logs (PII removed) plus synthetic edge cases.
- **LLM-as-judge**: open-ended outputs, scaling to thousands of test cases — but always validate judges against humans (target Spearman ≥ 0.8 , Cohen's $\kappa \geq 0.6$) before trusting.

Galileo's 2026 industry survey found ~74% of teams still use human-in-the-loop alongside automated; LLM-judge agreement with domain experts averages 64–68% in specialized domains. [Galileo AI](#)

6. Common pitfalls and best practices

The hardest lessons from 2024–2026 production deployments:

Benchmark contamination is real and pervasive. Sun et al. and Sainz et al. found 5–15% score inflation on public benchmarks; rephrasing benchmark questions causes large score drops. MMLU, HumanEval, GSM8K all show contamination on frontier models. Mitigate with post-cutoff benchmarks (LiveCodeBench, LiveBench), dynamic benchmarks (Chatbot Arena), or rephrased variants — and always hold out *uncontaminated* internal evals. (Meta Intelligence)

LLM-judge biases compound. Position bias (favors first option), length bias (prefers verbose), self-preference (judge prefers its own family), style-over-substance (coherent-but-wrong > less-coherent-but-faithful). Mitigations: randomize position, length-normalize, use multiple judges from different families, validate via Cronbach's α and Spearman vs humans. In one 2024 study, pairwise judges flipped ~35% of decisions under perturbation vs ~9% for absolute scoring.

Single-metric tunnel vision misleads. High faithfulness $\neq$ correctness if context is wrong. High pass@1 $\neq$ maintainable code. High BLEU $\neq$ usable translation. Always run a multi-metric scorecard weighted by business priority.

Saturation invalidates rankings. On HumanEval, frontier models hit 95%+; further gains are noise. Move to harder benchmarks (HumanEval+, BigCodeBench, HumanEval Pro, SWE-bench Verified, BBEH).

Implementation drift between harnesses produces conflicting scores. Hugging Face's MMLU study showed three implementations (LLaMA paper / EleutherAI / HELM) diverged by 10+ points for the same model. Always cite harness + version + prompt template.

The three-layer eval pyramid is the production pattern:

- **Tier 1 (continuous, cheap):** Heuristics, regex, JSON-schema validation, traditional metrics (F1, EM, ROUGE, BLEU)
- **Tier 2 (per PR/release):** LLM-as-judge on golden dataset (faithfulness, relevancy, safety)
- **Tier 3 (release / monthly):** Human review of sampled production traces, full red-team suite, benchmark replays

Operational discipline: version everything (dataset, prompt, model, harness); define thresholds tied to business KPIs (faithfulness ≥ 0.95 medical, ≥ 0.8 marketing); fail open on data quality, fail closed on safety; always pair "did it work" with "did it work safely"; evaluate retriever and generator separately for RAG. (deepeval)

7. GitHub repos and documentation

Framework	GitHub	Docs
RAGAS	github.com/explodinggradients/ragas	docs.ragas.io
DeepEval	github.com/confident-ai/deepeval	deepeval.com/docs
TruLens	github.com/truera/trulens	trulens.org
Giskard	github.com/Giskard-AI/giskard-oss	docs.giskard.ai
UpTrain	github.com/uptrain-ai/uptrain	docs.uptrain.ai
lm-evaluation-harness	github.com/EleutherAI/lm-evaluation-harness	in-repo docs/
OpenAI Evals	github.com/openai/evals	OpenAI Cookbook
OpenAI simple-evals	github.com/openai/simple-evals	README only
HELM	github.com/stanford-crfm/helm	crfm.stanford.edu/helm
BIG-Bench	github.com/google/BIG-bench	in-repo
MT-Bench	github.com/lm-sys/FastChat (fastchat/llm_judge/)	paper arXiv 2306.05685
AgentBench	github.com/THUDM/AgentBench	in-repo
EvalPlus	github.com/evalplus/evalplus	evalplus.github.io
HumanEval	github.com/openai/human-eval	paper arXiv 2107.03374
Promptfoo	github.com/promptfoo/promptfoo	promptfoo.dev/docs
MLflow	github.com/mlflow/mlflow	mlflow.org/docs/latest/genai/eval-monitor
Arize Phoenix	github.com/Arize-ai/phoenix	arize.com/docs/phoenix
W&B Weave	github.com/wandb/weave	weave-docs.wandb.ai
Inspect AI	github.com/UKGovernmentBEIS/inspect_ai	inspect.aisi.org.uk
Langfuse	github.com/langfuse/langfuse	langfuse.com/docs
Opik	github.com/comet-ml/opik	comet.com/docs/opik

Framework	GitHub	Docs
Braintrust Autoevals	github.com/braintrustdata/autoevals	braintrust.dev/docs
LangSmith openevals	github.com/langchain-ai/openevals	docs.smith.langchain.com
SWE-bench	github.com/SWE-bench/SWE-bench	swebench.com
GAIA	huggingface.co/gaia-benchmark	(HF dataset)
AlpacaEval	github.com/tatsu-lab/alpaca_eval	tatsu-lab.github.io/alpaca_eval
Arena-Hard-Auto	github.com/lmarena/arena-hard-auto	(paper arXiv 2406.11939)

Conclusion: what to take away

The field has converged on a clear pattern in 2026: combine reproducible public benchmarks for model selection, custom golden datasets for production gating, and LLM-as-judge for scaling open-ended evaluation — never trust any single metric or framework in isolation. The LLM-as-judge methodology, pioneered by MT-Bench and refined by G-Eval/Prometheus/QAG decomposition, has become the workhorse for any criterion lacking ground truth, but it inherits judge biases that must be calibrated against humans before trusting in production.

Three structural shifts deserve attention. First, agent evaluation has become the new frontier — outcome metrics alone hide unsafe trajectories, so SWE-bench Verified, GAIA2, AgentProcessBench, and Inspect AI's sandboxed framework are reshaping what "good" looks like. Second, benchmark contamination has invalidated naive leaderboard reporting; serious teams now hold out private evals and use post-cutoff benchmarks like LiveBench. Third, the OSS observability tier (Langfuse, Phoenix, Opik) has caught up with closed-source SaaS on capabilities while remaining cost-predictable at scale — making the open vs managed decision genuinely competitive in 2026, not merely an OSS-purist preference.

The pragmatic recommendation for most teams shipping LLM products: start with DeepEval (Pytest-style local CI) plus Ragas (if RAG-heavy), graduate to Langfuse or Phoenix for production observability, run Promptfoo or DeepTeam before every model upgrade for red-team coverage, and reach for lm-evaluation-harness only when you need reproducible benchmark scores. Pair every metric with a domain-specific custom golden dataset — that single artifact, more than any framework choice, separates teams that catch regressions before customers from teams that don't.